Dylan Becchetti

Mechanical Engineering | Systems, Automation & Engineering Software

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Portfolio: formsby-systems.github.io/engineering-portfolio

ENGINEERING EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024-2025 | Engineering Intern, summer 2023

- Built a self-directed Excel/VBA production tracker used by engineers and operators in daily R&D shift debriefs.
- Automated weekly Scorecard defect counts from 10,000+ unit records and connected production, yield and delivery reporting; developed date-based calculations and downtime allocation.
- Linked purchasing and inventory records in Smartsheet, reconciled part identifiers, and planned spare-parts storage and signage across 14 locations.
- Wrote illustrated sample-build instructions, measured cleanroom airflow with a hood and micromanometer, analyzed data in Minitab and drafted AutoCAD layouts.

SELECTED PROJECTS

Kinetic Wilbur | Mechanical Design, year-long team capstone

- Designed and iterated the full-scale OpenSCAD assembly for an 80-inch suspended planetary-gear sculpture, translating stakeholder needs and sponsor feedback into manufacturable parts.
- Modeled welded-steel carrier arms and cable stays, revised ring joints and drive alignment, and prepared CNC profiles; worked with the team on FEA, fit checks and materials selection.

Midas Futures | Creator & Technical Lead

- Created a Python futures research and Practice platform with modular strategies, batch campaigns, component ablations, Monte Carlo studies and account-lifecycle simulation.
- Matched 228 session decisions and 225 modeled trade lifecycles against a frozen reference. A separate cloud controller resumed from a checkpoint to finish 36 account simulations and preserve 216 prior results.

Operalith | Founder & Product Lead, independent product

- Built a configurable React/Next.js workspace linking customer enquiries, projects, tasks and supplies through shared records and compatible tool features.
- Implemented reciprocal enquiry/project links and targeted undo; designed PostgreSQL saves with ownership, revision and request-identity checks for stale edits and retries.

Formsby | Creator, cross-source search and matching

- Developed Python discovery and later Next.js/PostgreSQL housing-search prototypes, with queued source adapters, URL deduplication, scored listing evidence and private results tied to the owner's latest run.

Scanned Leverage and Optimized Production | Creator

- Built Python/Luau Roblox market-research and production tooling to structure gameplay observations and assemble original prototypes; reproduced two configuration-driven builds byte for byte.

EDUCATION

Lehigh University | B.S. Mechanical Engineering, expected January 2027

Materials Science and Engineering minor | Coursework expected December 2026 | CSWA, December 2024

TECHNICAL SKILLS

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Smartsheet, Minitab.